TING-LUN SU

R&D Engineer | AI & Robotics Specialist

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PROFESSIONAL SUMMARY

Research-oriented R&D engineer with expertise in **AI-driven robotics and intelligent manufacturing**. Led 6 industry-academia projects deploying 30+ automated systems (20 CNC machines, 10 robotic arms) achieving **80% manpower reduction** and **91% defect reduction**. Published 2 journal papers (1 SCI-indexed Q2) and holds 2 patents. Specialized in computer vision, path optimization algorithms, and large-scale automation system integration for Industry 4.0 applications.

EDUCATION

B.S. in Intelligent Automation Engineering | Expected: June 2026
National Chin-Yi University of Technology (NCUT), Taichung, Taiwan

• Focus: AI-driven robotics, intelligent manufacturing, computer vision, industrial automation
• Led multiple large-scale industry-academia collaborative research projects
• Active contributor to academic publications and patent development

TECHNICAL SKILLS

Programming: Python, C#, C++, MATLAB
AI & Machine Learning: YOLO Object Detection, Computer Vision (OpenCV), Deep Learning, Algorithm Development
Robotics & Automation: Robot Arm Programming & Control, Path Planning (Ant Colony Optimization, Bayesian Optimization), System Integration, PLC Programming, CNC Communication
CAD/CAM: SolidWorks, 3D Modeling, Mechanical Design
Emerging Tech: Large Language Models (LLM), Model Context Protocol (MCP), Generative AI Applications
Industrial Systems: Industrial Control Systems, Smart Manufacturing, Industry 4.0 Solutions

PROFESSIONAL EXPERIENCE

Lead R&D Engineer | Visual Measurement & AI Navigation System | **Jan 2025 – Present**
TURVO INTERNATIONAL CO., LTD., Taichung, Taiwan

• Lead project planning, technical architecture design, and stakeholder communication for AI-powered quality inspection system
• Developed intelligent navigation assistance software integrating LLM and Model Context Protocol (MCP)
• Designed computer vision-based measurement inspection station with real-time defect detection capabilities
• Implemented user guidance system leveraging generative AI for operational efficiency improvement

Project Lead | Smart CNC Flexible Manufacturing Factory (3-Phase) | **Jan 2024 – Dec 2025**
IC STAR INDUSTRIAL CO., LTD., Taichung, Taiwan

• Led 42-member cross-functional team through complete factory automation transformation across two plants
• Deployed 20 CNC machines and 10 robotic arms with full hardware-software-control system integration
• **Impact:** Achieved **80% manpower reduction**, significantly enhanced production throughput and operational efficiency
• Managed end-to-end implementation: robotics programming, CNC communication protocols, HMI design, control system architecture
• Redesigned Plant A layout and constructed new Plant B with Industry 4.0 standards

AI Algorithm Lead | Automated Forging Spraying System (2-Phase) | **Jun 2023 – Dec 2024**
IC STAR INDUSTRIAL CO., LTD., Changhua, Taiwan

• Developed AI-based path planning using Ant Colony Optimization and Bayesian Optimization for robotic spray coating
• **Results: 27% production increase, 91% defect reduction, 21% cost savings** across 3 automated forging lines

• Achieved 97.3% trajectory matching accuracy and 83% reduction in production line switching time
• Integrated robotic arm control with real-time process monitoring and adaptive path adjustment

Computer Vision Engineer | CNC Chip Entanglement Detection System | Mar 2023 – Aug 2023

Industrial Technology Research Institute (ITRI), Taichung, Taiwan

• Developed real-time AI-based YOLO vision system for detecting CNC tool chip entanglement in unmanned lines
• Reduced machine downtime and maintenance costs through early fault detection and prevention
• Deployed solution in smart manufacturing environment with 24/7 operational requirements

Product Lead | Automatic Mushroom Stem-Cutting System | Sep 2022 – Present

University Social Responsibility (USR) Project, Xinshe, Taiwan

• Led product development from prototype to commercialization, achieving mass production adopted by 3 manufacturers
• Designed and patented innovative pressing mechanism and auto-sorting system (2 patents granted)
• Improved agricultural processing efficiency and product quality consistency through automation
• Managed full product lifecycle: mechanical design, control system, field testing, client deployment

System Integration Engineer | 3D Vision-Guided Robotic Picking | Sep 2022 – Present

LIEBHERR-Verzahntechnik GmbH & PONGI Gear Expert CO., LTD.

• Integrated and optimized Liebherr LHRobotics.Vision random picking system for Taiwan market deployment
• Developed 3D vision-guided robotic coordination algorithms for enhanced precision and flexibility
• Provided technical consulting for client implementation, troubleshooting, and performance optimization

PUBLICATIONS

Journal Articles

T.-L. Su, P.-J. Chen, T.-J. Peng, and C.-C. Lin, "Automated spray path planning based on Bayesian optimization and ant colony optimization," *The International Journal of Advanced Manufacturing Technology*, vol. 139, pp. 5491-5509, Aug. 2025.
DOI: 10.1007/s00170-025-16162-x | **SCI-indexed, Q2, Impact Factor: 2.9**

T.-J. Peng, P.-J. Chen, **T.-L. Su**, and C.-C. Lin, "Optimization of Automated Forging Release Agent Spray Path Based on Ant Colony Algorithm," *Journal of Automation Intelligence and Robotics*, TAIROA, No. 51, pp. 114-120, Dec. 2024.

Selected Conference Papers

• **T.-L. Su**, C.-H. Du, and C.-H. Lai, "Development of a Contactless AIoT Smart Data Management Hybrid Vending System," *17th Intelligent Living Technology Conference*, Jun. 2023.
• Z.-L. Sun et al. (incl. **T.-L. Su**), "Green Machining Optimization and Burr Treatment in Mill-Turn Operations," *27th CSMMT Annual Meeting*, Nov. 2024.

PATENTS

• **Invention Patent:** Mushroom Stem Cutting Machine | Taiwan Patent No. **I891383** (Granted May 2025)
Automated agricultural processing device with precision cutting mechanism
• **Utility Model Patent:** Mushroom Stem Cutting Machine | Taiwan Patent No. **M659734** (Granted Aug 2024)
Enhanced mechanical design for improved stability and cutting performance

HONORS & AWARDS

• **2nd Place** | National Vocational and Technical College Student Practical Project Competition (2025)
• **1st Place** | Intelligent Automation Equipment Invention Awards, 13th Edition (2024)
• **3rd Place** | Taiwan Society of Precision Engineering's 113th Project and Paper Award (2024)
• **Honorable Mention** | Mitsubishi Electric Contest of Automation, 1st Edition (2024)
• **Science and Technology Humanities Award** | 27th TDK Cup National Collegiate Creative Design (2023)
• **Early Achievements (2020-2022):** 3 Gold, 2 Silver, 4 Honorable Mentions in National Competitions

CONFERENCE PRESENTATIONS & INDUSTRY EXHIBITIONS

• **2025 Taipei USR EXPO** (Sep 2025) – Presented sustainable agricultural automation with AI-based control
• **2024 Taipei International Automation Exhibition** (Aug 2024) – Showcased AI robotic spraying system, initiated collaboration

PROFESSIONAL CERTIFICATIONS

• Photovoltaic System Installation (Class B) – Ministry of Labor (WDA)
• AI Application Planner (iPAS Certified) – Ministry of Economic Affairs
• Industrial Wiring (Class C) & Commercial Wiring (Class C) – Ministry of Labor (WDA)

INTERNATIONAL ACADEMIC EXCHANGE

SAKURA SCIENCE Exchange Program | Japan Science and Technology Agency (JST) | 2024
Suwa University of Science, Nagano, Japan | Host: Prof. Masahide Oshima (Vice President)

LANGUAGES

• **Chinese:** Native proficiency
• **English:** Professional working proficiency (TOEIC: 660)
• **German:** Elementary proficiency

References available upon request